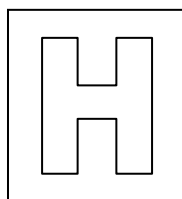


Candidate Name: _____

Class Adm No

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2011 Promotional Examination II Pre-university 2

H1 CHEMISTRY

8872 / 02

Paper 2

15 September 2011

2 hours

Additional Materials: Cover page
Data Booklet
5 pieces of writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, index number and class on all the work you hand in.

Write in dark blue or black pen on both sides of the writing paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluids.

Section A

Answer **all** questions.

Section B

Answer any **two** questions on the writing papers. Start each new question on a fresh sheet of writing paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of M_r and A_r , unless a different level of accuracy is specified in the question.

You are reminded of the need for clear presentation in your answers and to show all working in calculations.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

	Section A			Section B			Total
Question No	1 (15m)	2 (10m)	3 (15m)	4 (20m)	5 (20m)	6 (20m)	80
Marks Obtained							

This question paper consists of # printed pages.

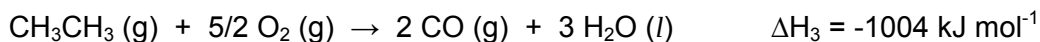
[Turn over

SECTION A (40 marks)

Answer all questions in the spaces provided.

1 The main compounds discussed in this question are ethane and ethene.

- (a)** When burnt in a limited supply of air, ethane forms carbon monoxide and water according to the reaction shown below.

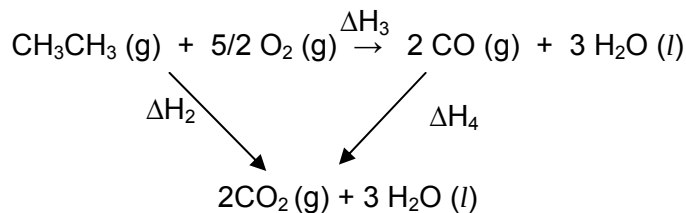


Ethane can be synthesized through the hydrogenation of ethene as shown in the following equation: $\text{CH}_2=\text{CH}_2 (\text{g}) + \text{H}_2 (\text{g}) \rightarrow \text{CH}_3\text{CH}_3 (\text{g}) \quad \text{-----} \Delta H_1$

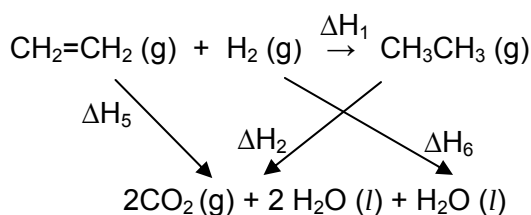
Given the following energy cycles and various enthalpy change data,

$\text{CH}_3\text{CH}_3 (\text{g}) + 5/2 \text{O}_2 (\text{g}) \rightarrow 2 \text{CO} (\text{g}) + 3 \text{H}_2\text{O} (\text{l})$	$\Delta H_3 = -1004 \text{ kJ mol}^{-1}$
Enthalpy change of combustion of $\text{CH}_2=\text{CH}_2 (\text{g})$	$\Delta H = -1411 \text{ kJ mol}^{-1}$
Enthalpy change of formation of $\text{H}_2\text{O} (\text{l})$	$\Delta H = -286 \text{ kJ mol}^{-1}$
Enthalpy change of combustion of $\text{CO} (\text{g})$	$\Delta H = -283 \text{ kJ mol}^{-1}$

Energy Cycle 1



Energy Cycle 2



- (i)** Identify the enthalpy changes represented by ΔH_4 , ΔH_5 , and ΔH_6 .

ΔH_4 : Enthalpy change of combustion of $\text{CO} (\text{g})$

ΔH_5 : Enthalpy change of combustion of $\text{CH}_2=\text{CH}_2 (\text{g})$

ΔH_6 : Enthalpy change of formation of $\text{H}_2\text{O} (\text{l})$

;

- (ii) Use both energy cycles and the enthalpy change data given to calculate the enthalpy change of hydrogenation of ethene, ΔH_1 .

From energy cycle 1,

Enthalpy change of combustion of ethane = $-1004 + (2 \times -283) = -1570 \text{ kJ mol}^{-1}$

From energy cycle 2,

$\Delta H_1 = -1411 + (-286) - (-1570) = \underline{-127 \text{ kJ mol}^{-1}}$

Allow e.c.f. $\Delta H_c (\text{C}_2\text{H}_6)$

; correct application of Hess' Law

; ; final answer with units.

- (iii) Define the term *Standard Enthalpy Change of Combustion*, ΔH_c^θ .

.....
.....

Enthalpy change when one mole of a substance is completely burnt in excess oxygen under standard conditions of 1 atm and 298K.

;
;

- (iv) The carbon-oxygen bond energy of carbon monoxide is $+1074 \text{ kJ mol}^{-1}$ while the carbon-oxygen bond energy of carbon dioxide is $+802.5 \text{ kJ mol}^{-1}$. Why is the carbon-oxygen bond energy in carbon monoxide different from that in carbon dioxide?

.....
.....

The carbon-oxygen bond in CO is a triple bond whereas that in CO₂ is a double bond; more electrons shared between C and O in CO, stronger electrostatic attraction between C and O nuclei and electrons in CO, so carbon-oxygen bond in CO is stronger.

;

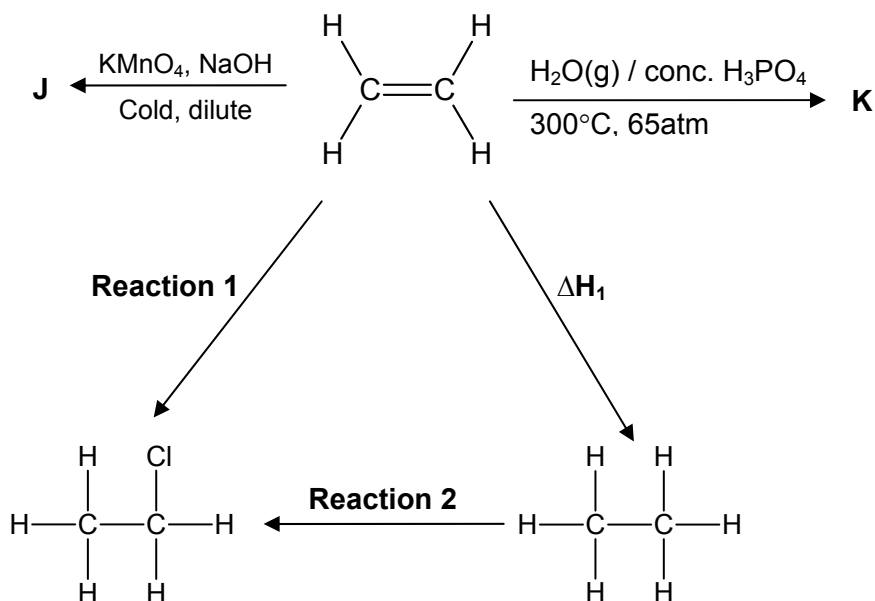
- (v) State one environmental consequence of excessive production of carbon monoxide arising from the combustion of car engines.

.....
CO will bond with haemoglobin, hence, reducing the ability of blood to carry oxygen round the body, resulting in poisoning.

;

[8]

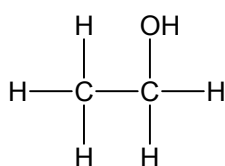
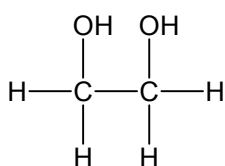
(b) Consider the following reaction scheme:



(i) Draw the structure of the organic products **J** and **K**.

J

K



(ii) State suitable reagents and conditions required for:

Reaction 1

Reagents: HCl (g)

Conditions: room temperature

Reaction 2

Reagents: $\text{Cl}_2(\text{g})$, sunlight/UV

Conditions: high temperature

[4]

- (c) Group I and Group II metals when burnt in air produces metal oxides.

The melting points of two such oxides are given below.

	Na ₂ O	MgO
melting point/ °C	1132	2852

- (i) Explain the difference in their melting points.

.....
.....
.....
.....
.....
.....

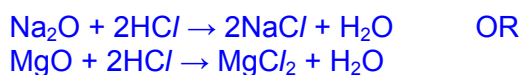
Both Na₂O and MgO are ionic compounds with strong electrostatic forces of attraction between oppositely charged cations. However, the magnitude of the lattice energy of Na₂O is smaller than MgO.

This is because, the magnitude of lattice energy ($\propto \frac{q^+q^-}{r^+ + r^-}$). Since Mg²⁺ has twice the charge and has a smaller cationic radius, MgO has a larger lattice energy, thus more energy is required to overcome the strong electrostatic forces.

- (ii) Write an equation to describe how **any one** of these oxides reacts with dilute hydrochloric acid.

.....

[3]



[Total: 15]

2

Elemental nitrogen is a colorless, odorless and mostly inert diatomic gas at standard conditions, constituting 78.08% by volume of the Earth's atmosphere. Nitrogen has the symbol **N**, atomic number 7 and atomic mass 14.00674. Natural nitrogen consists of two stable isotopes, ^{14}N , which makes up the vast majority of naturally occurring nitrogen, and ^{15}N .

Nitrogen readily forms oxides of nitrogen when elemental nitrogen is combusted at high temperatures. Examples of oxides of nitrogen include N_2O_3 , N_2O_4 , and N_2O_5 , N_2O , NO_2 , and NO .

N_2O_3 , N_2O_4 , and N_2O_5 decompose rapidly at room temperature, N_2O is stable and rather unreactive at room temperature, while NO and NO_2 are quite reactive but stable when isolated.

Out of all these nitrogen containing compounds, N_2O_4 is one of the most important rocket propellants ever developed as it burns on contact without a separate ignition source, making it a powerful oxidizer.

- (a) Calculate the percentage abundance of the ^{15}N isotope.

[2]

- (a) Let the % abundance of ^{14}N be x . Therefore, % abundance of ^{15}N is $(100-x)$.

$$14.00674 = \frac{14x}{100} + \frac{15(100-x)}{100}$$

;

$$x = 99.326$$

Therefore, % abundance of $^{15}\text{N} = 100 - 99.326 = \underline{0.674\% (3s.f)}$

;

- (b) (i) State the electronic configuration of the elements N and O.

N: $1s^2 2s^2 2p^3$

O: $1s^2 2s^2 2p^4$

;

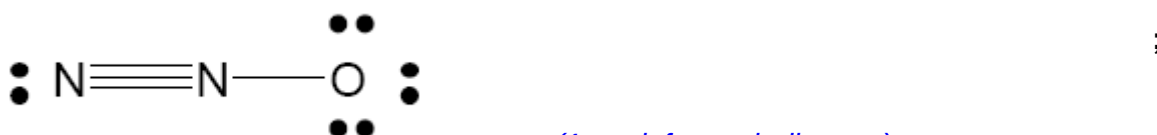
- (ii) Using the electronic configuration written in (b)(i), explain if the first ionization energy of N is higher or lower than O.

[2]

The first ionization energy of N is higher than O. There is mutual repulsion between the paired electrons in the 2p orbitals in O, resulting in the electron to be easily removed.

- (c) By drawing suitable 'dot-and-cross' diagrams, explain the following statement in *italics*,
 (i) '*N₂O is stable and rather unreactive at room temperature, while NO and NO₂ are quite reactive*'.

(*Hint: NO and NO₂ have structures that do not follow the octet rule)



(1 mark for each diagram)

Since NO and NO₂ contains a lone unpaired electron, it behaves like a radical and it is very chemically reactive as it wants to react to achieve octet configuration.

- (ii) Compare the boiling points of N₂ and NO₂. Which of these two compounds has a higher boiling point? Explain your answer.

[6]

Both compounds are simple covalent molecules. NO₂ forms permanent-dipole forces of attraction between their molecules while N₂ formed temporary dipole-induced dipole forces of attraction between their molecules.

Since pd-pd is stronger than td-id, NO₂ will have a higher boiling point, thus requiring a greater amount of heat to overcome these forces.

[Total: 10]

- 3 (a) The initial rate of reaction between compound **A**, HCOOCH₂CH₂CH₃, and aqueous sodium hydroxide was measured in a series of experiments at a constant temperature. The data obtained are shown below:

Expt.	Initial concentration of NaOH / mol dm ⁻³	Initial concentration of A / mol dm ⁻³	Initial rate / mol dm ⁻³ s ⁻¹
1	0.040	0.030	4.0 × 10 ⁻⁴
2	0.040	0.045	6.0 × 10 ⁻⁴
3	0.060	0.045	9.0 × 10 ⁻⁴
4	0.120	0.060	b

- (i) Use the data in the table to deduce the order of reaction with respect to **A** and the order of reaction with respect to NaOH.
 (ii) Hence write a rate equation for this experiment.

(iii) Calculate the initial rate of reaction, **b**, in Experiment 4.

[5]

(i) By comparing experiments 1 & 2, when [NaOH] is kept constant and [A] increased by a factor of 1.5, the rate of reaction also increased by a factor of 1.5 Thus the order of reaction is first order with respect to A.

By comparing experiments 2 & 3, when [A] is kept constant and [NaOH] increased by a factor of 1.5, the rate of reaction also increased by a factor of 1.5 Thus the order of reaction is first order with respect to NaOH.

(ii) Rate = $k [A] [NaOH]$

(iii) From experiment 1,

Rate = $k [A] [NaOH]$

$$4.0 \times 10^{-4} = k (0.030) (0.040)$$

$$0.3333 = k$$

From experiment 4,

Rate = $k [A] [NaOH]$

$$\mathbf{b} = 0.3333 (0.060) (0.120) = \underline{0.00240 \text{ mol dm}^{-3} \text{ s}^{-1}} \text{ (3.s.f)}$$

(b) In a further experiment at a different temperature, the initial rate of reaction was found to be $9.0 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$ when the initial concentration of **A** was $0.020 \text{ mol dm}^{-3}$ and the initial concentration of NaOH was 2.00 mol dm^{-3} .

Under these new conditions, the reaction is first order with respect to **A** and appears to be zero order with respect to sodium hydroxide.

(i) Write a rate equation for the reaction under these new conditions.

$$\text{Rate} = k_1 [A]^1 [NaOH]^0 \Rightarrow \text{Rate} = k_1 [A]^1$$

(ii) Calculate a value for the rate constant, k' , under these new conditions and state its units.

[3]

To solve for k_1 ,

$$9.0 \times 10^{-3} = k_1 (0.020)$$

$$k_1 = \underline{0.450 \text{ s}^{-1}}$$

(c) Compounds **P** and **Q** have the molecular formula C_3H_8O .

Both **P** and **Q** decolourises hot acidified $KMnO_4$ to form **R** and **S** respectively.

R reacts with Na_2CO_3 to produce an acidic gas but **S** does not.

S forms a crystalline orange precipitate with 2,4-dinitrophenylhydrazine but does not form a silver mirror with Tollen's reagent.

Draw one possible structure for **P**, **Q**, **R** and **S**, explaining your deductions clearly.

<u>Observations</u>	<u>Deductions</u>
Both P and Q decolourises hot acidified $KMnO_4$ to form R and S respectively.	P and Q contains either a 1° or 2° alcohol functional group ;
R reacts with Na_2CO_3 to produce an acidic gas but S does not.	R contains a <u>carboxylic acid</u> functional group. The acidic gas produced is CO_2 ;
S forms a crystalline orange precipitate with 2,4-dinitrophenylhydrazine but does not form a silver mirror with Tollen's reagent.	S contains a <u>ketone</u> functional group ;

Structure of P	Structure of Q
$CH_3CH_2CH_2OH$;	$CH_3CH(OH)CH_3$;
Structure of R	Structure of S
CH_3CH_2COOH ;	CH_3COCH_3 ;

[7]

[Total: 15]

SECTION B (40 marks)

Answer only 2 out of the 3 questions

- 4 (a) Ammonia is manufactured from nitrogen and hydrogen by the Haber Process.



- (i) Apart from its use in making fertilizers, give another reason why the manufacture of ammonia is important.

Ammonia is important in the manufacturing of chemicals such as nitric acid and urea
OR

Ammonia is also used as a cleaning agent to remove grease.

;

- (ii) Explain why the activation energy of the Haber Process is high.

Activation energy is high due to the breaking of the very strong $\text{N}\equiv\text{N}$ triple bond.

;

- (iii) State Le Chatelier's Principle.

Le Chatelier's principle states that when a system in equilibrium is disturbed, the system will react to counteract the change imposed so as to re-establish the equilibrium.

;

;

- (iv) Explain why the Haber Process is carried out under high pressure and low temperature conditions.

By LCP, under conditions of high pressure, the system will counteract by reducing the pressure, thus the equilibrium will shift to the right where they are lesser number of gaseous particles to reduce the pressure. Hence increasing the yield of ammonia.

;;

Under conditions of low temperature, by LCP, the system will counteract by increasing temperature, and since forward reaction is exothermic, equilibrium will shift to the right, thereby producing more heat and also increasing the yield of ammonia.

;;

Under conditions of high pressure and low temperature, the yield of ammonia will increase.

- (v) Write an expression for K_c , and hence calculate the value of K_c given the following equilibrium concentrations at 1000K. State the units of K_c .

Gas	Concentration / mol dm^{-3}
N_2	1.36
H_2	1.84
NH_3	0.142

[11]

$$K_c = \frac{(0.142)^2}{(1.84)^3 \times 1.36}$$

$$= \underline{0.00238 \text{ mol}^{-2} \text{ dm}^6}$$

;

;;

(b) Consider the organic compound with the structural formula of $\text{CH}_3\text{CH}=\text{CH}(\text{OH})$.

(i) Define the term *structural isomers*.

(ii) Draw **and** name the structures of two isomers of $\text{CH}_3\text{CH}=\text{CH}(\text{OH})$ that have **different** functional groups.

[4]

(c) Halogenoalkanes are useful in the laboratory as intermediates in making other organic compounds.

(ii) State in increasing order, the reactivity of iodoethane, bromoethane and chloroethane. Explain your answer.

[2]

(d) Butan-2-ol, $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$, can undergo dehydration to form two products which are positional isomers of each other.

(i) Suggest the reagents and conditions for the dehydration.

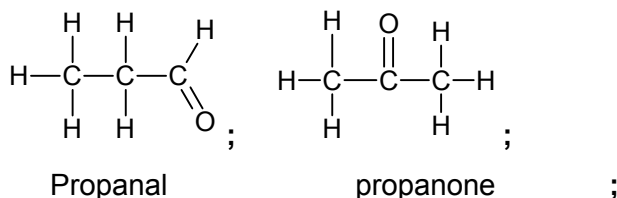
(ii) Draw the structural formula of the two products.

[3]

[Total: 20]

(b) (i) Structural isomers are compounds with the **same molecular formula** but **different structural formula** ;

(ii) Draw **and** name the structures of two isomers of $\text{CH}_3\text{CH}=\text{CH}(\text{OH})$ that have **different** functional groups.



(c) State the order of reactivity of iodoethane, bromoethane and chloroethane. Explain your answer.

The reactivity of the alkyl halides increases in the order:



This is due to the **size** of the halogen atom (going down the group Cl, Br, I: size increases). The **bigger** the atom, the **weaker** the C-X bond. Hence the **easier** it is to break the bond leading to **higher** reactivity (or faster reaction) ;

(d) Butan-2-ol, $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$, can undergo dehydration to form two products which are positional isomers of each other.

(i) Suggest the reagents and conditions for the dehydration.

Excess conc. $\text{H}_2\text{SO}_4(\text{aq})$ at 180°C ;

(ii) Draw the structural formula of the two products.

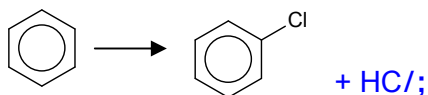
$\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$; $\text{CH}_3\text{CH}=\text{CHCH}_3$;

5 (a) Benzene can react with chlorine to form chlorobenzene.

(i) What are the reagents and conditions needed for this reaction to occur?

Cl_2 in the presence of FeCl_3 or AlCl_3 catalyst ;

(ii) Write a balanced equation for this reaction.



(iii) Explain why benzene undergoes substitution reactions but not addition reactions.

Benzene consists of a delocalized e^- cloud above and below the plane of the benzene ring which stabilizes the ring structure of benzene. Benzene undergoes electrophilic substitution reactions as the aromatic character not destroyed, i.e. maintaining of the delocalised π electron cloud. Benzene is resistant to addition reactions as these would destroy the delocalized π electron cloud ;

[3]

(b) A sequence of reactions, starting from ethanal, is shown below.



(i) Suggest the reagents and conditions for steps, I and II.

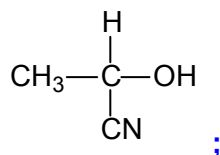
I : KCN , $\text{H}_2\text{SO}_4(\text{aq})$ at $10 - 20^\circ\text{C}$;

II : dilute H^+ , reflux ;

(ii) Name the type of reaction for step I.

Addition ;

(iii) Draw the structural formula of T.



[4]

(c) Esters of carboxylic acids are often used as components in fruit flavouring. Ethyl butanoate, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_3$, is a major component of strawberry and pineapple flavouring.

What are the reagents and conditions to synthesise ethyl butanoate?

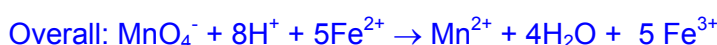
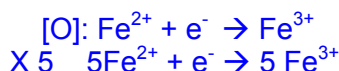
$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$; $\text{CH}_3\text{CH}_2\text{OH}$; conc H_2SO_4 , reflux ;

[3]

- (d) Potassium manganate is an oxidising agent commonly used in organic synthesis.

FA 1 contains Fe^{2+} ions. In an experiment, it was found that 25.0 cm^3 of **FA 1** required 18.00 cm^3 of $0.200 \text{ mol dm}^{-3}$ acidified KMnO_4 (aq) for complete reaction.

- (i) With reference to the *Data Booklet*, state the oxidation and reduction half-equations for this reaction.
- (ii) Hence, write an overall equation (with state symbols) for this reaction.
- (iii) Calculate the amount of Fe^{2+} present in 25.0 cm^3 of **FA 1**.



Hence, mol. ratio of $\text{Fe}^{2+} : \text{MnO}_4^- = 5:1$

$$\text{Amount of } \text{MnO}_4^- \text{ used} = \frac{18.00}{1000} \times 0.200 = 3.600 \times 10^{-3} \text{ mol.}$$

$$\text{Amount of } \text{Fe}^{2+} \text{ present in } 25.0 \text{ cm}^3 = 5 \times 3.600 \times 10^{-3} = 0.01800 \text{ mol.}$$

- (e) **Isoelectronic** atoms or ions have the same number of electrons. Those with the same number of neutrons are **isotonic**, and those with the same number of protons are **isotopic**.

- (i) Given the following ions, ${}_{15}^{31}\text{V}^{3-}$, ${}_{15}^{32}\text{W}^{2-}$, ${}_{16}^{31}\text{X}^{2-}$, ${}_{15}^{31}\text{Y}^{-}$.

Which of the following ions are:

- (I) Isoelectronic (Answer: V, X)
- (II) Isotonic (Answer: V and Y)
- (III) Isotopic (Answer: V, W and Y)

- (ii) Which of the ion will be

- (I) deflected the most, (Answer: V)
- (II) deflected the least (Answer: Y)

when passed through an electric field?

[5]

[Total: 20]

- 6 (a) Acetic acid is an organic compound with the chemical formula CH_3COOH . It is a colourless liquid that when undiluted is also called glacial acetic acid. It is a cigarette additive and, as the main component of vinegar, has a distinctive sour taste and pungent smell.

The acid dissociation constant K_a of ethanoic acid is $1.8 \times 10^{-5} \text{ mol dm}^{-3}$.

- (i) Write an expression for the K_a of ethanoic acid.

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]} \quad ;$$

- (ii) Given that $[\text{H}^+] = \sqrt{[\text{weak acid}]K_a}$, calculate the $[\text{H}^+]$ and the pH of a 0.10 mol dm^{-3} solution of ethanoic acid.

$$[\text{H}^+] = [(1.8 \times 10^{-5})(0.10)]^{1/2} \\ = \underline{0.00134 \text{ mol dm}^{-3}} \quad ;$$

$$\text{pH} = -\lg(0.00134) = \underline{2.87} \quad ;$$

- (iii) Explain why ethanoic acid is a weak acid.

[4]

Ethanoic acid is a weak acid as $[\text{H}^+] \ll [\text{CH}_3\text{COOH}]$. This shows that ethanoic acid only partially dissociates in solution. ;

- (b) A solution containing ethanoic acid and its salt, sodium ethanoate, $\text{CH}_3\text{CO}_2\text{Na}$, acts as a buffer solution.

- (i) What does the term *buffer solution* mean?

A buffer solution is a solution whose pH remains almost unchanged if a little acid or alkali is added to it. ;

- (ii) Describe with equations how a buffer solution works when a small amount of H^+ ions and OH^- ions are added to it.

[3]

When OH^- is added to the solution,



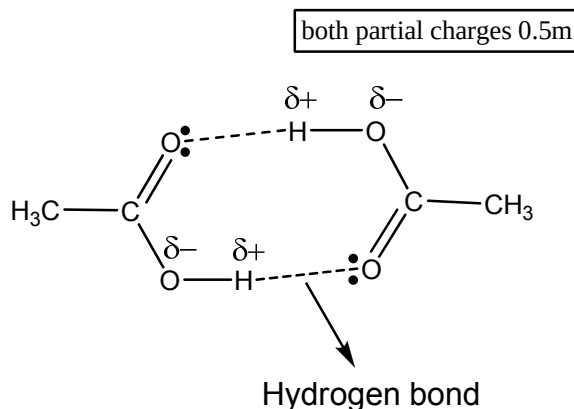
When H^+ is added to the solution,



- (c) The relative molecular mass of ethanoic acid is 60. However, when ethanoic acid is dissolved in benzene, it is found that the relative molecular mass of the acid is 120. Explain this phenomenon using a clearly labeled diagram.

[2]

In benzene, two molecules of ethanoic acid combine together to form a dimer through hydrogen bonding. ;



- (d) The standard enthalpy of neutralisation can be determined when solutions of an acid and a base are mixed. The container used for measuring the change in temperature is called a calorimeter.

- (i) 250 cm³ of sodium hydroxide solution of concentration 0.400 mol dm⁻³ was added to 250 cm³ of ethanoic acid of concentration 0.400 mol dm⁻³ in a calorimeter. The initial temperature of the two solutions was 17.05 °C. At the end of the reaction, the temperature rose to 19.55 °C.

Assuming the specific heat capacity of all the solutions is 4.2 J g⁻¹ K⁻¹, calculate the standard enthalpy of neutralisation **per mole of water formed**.

$$Q = mc\Delta T = 500 \times 4.2 \times (19.55 - 17.05) = 5250 \text{ J} \quad ;$$

$$\text{Amount of water} = 250/1000 \times 0.4 = 0.100 \text{ mol} \quad ;$$

$$\text{enthalpy of neutralisation} = - 5250 / 0.100 = - 52.5 \text{ kJ mol}^{-1}$$

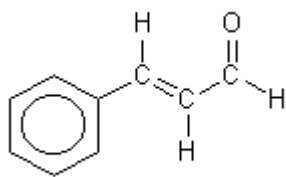
; final answer ; units and sign

- (ii) State one possible source of error in this experiment.

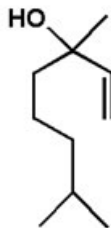
Heat loss to the surroundings, thus the highest temperature recorded may not be accurate. ;

[5]

- (e) The following compounds are two of the many chemicals present in cinnamon.



A



B

- (i) Describe **two** chemical tests that can be carried out to distinguish between both compounds. You should also state observations for each of the tests in your answers.

2-4 DNPH. Orange ppt observed with **A**, no ppt observed for **B**.

;;

OR

;;

Tollen's reagent. Silver mirror observed with **A**, no silver mirror for **B**.

OR

$K_2Cr_2O_7/H^+$, heat. Orange solution of $K_2Cr_2O_7$ turns green in **A**, Orange colour remains in **B**.

- (ii) Describe one chemical test that can give the same positive result when carried out on both compounds. State the observations that you will see.

Br_2 (aq), warm.

;;

Reddish brown bromine decolourise.

[6]

[Total: 20]

End of Paper